



# Time Series Analysis

## Chapter 3: Unit roots and ARIMA models



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## Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AIC**: Akaike Information Criterion
- ▣ **AR**: AutoRegressive (model)
- ▣ **ARIMA**: AutoRegressive Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CUSUM**: CUmulative SUM (control chart)
- ▣ **DF**: Dickey–Fuller (test)
- ▣ **DF-GLS**: Dickey–Fuller test with GLS detrending
- ▣ **DS**: Difference–Stationary (series)
- ▣ **DW**: Durbin–Watson (statistic)
- ▣ **EU**: European Union
- ▣ **FRED**: Federal Reserve Economic Data

## Acronyms used in this material (2/2)

- **GDP:** Gross Domestic Product
- **GLS:** Generalised Least Squares
- **GNP:** Gross National Product
- **HICP:** Harmonised Index of Consumer Prices
- **KPSS:** Kwiatkowski–Phillips–Schmidt–Shin (test)
- **MA:** Moving Average
- **MAIC:** Modified Akaike Information Criterion
- **OLS:** Ordinary Least Squares
- **PACF:** Partial AutoCorrelation Function
- **PP:** Phillips–Perron (test)
- **RMSE:** Root Mean Squared Error
- **SARIMA:** Seasonal ARIMA (model)
- **SE:** Standard Error
- **TBATS:** Trigonometric seasonality, Box–Cox, ARMA errors, Trend, Seasonal components (model)
- **TS:** Trend-Stationary (series)
- **US:** United States
- **VAR:** Vector AutoRegression
- **ZA:** Zivot–Andrews (test)

## Today's question and route

- **Question:** when a series drifts upwards, is the shock of today temporary or permanent, and how should we forecast it?
  - ▶ the answer decides whether we detrend or difference, and how wide our forecast intervals are
- **Route** of the chapter
  - ▶ deterministic and stochastic trends; integrated processes  $I(d)$ ; spurious regression
  - ▶ unit-root tests: Dickey–Fuller, ADF, Phillips–Perron; the KPSS stationarity test; structural breaks
  - ▶ ARIMA( $p, d, q$ ) models: identification, estimation, diagnostics, forecasts with growing intervals
- We build on [Chapter 1](#) (stationarity, random walk, differencing) and [Chapter 2](#) (ARMA models); Seminar 3 comes before this lecture

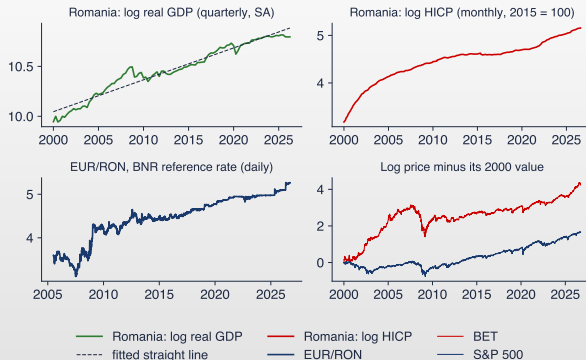
## Learning outcomes

- ▣ Distinguish trend-stationary from difference-stationary series, and explain why the distinction matters
- ▣ Recognise a spurious regression and explain why it happens
- ▣ Run and interpret the ADF, Phillips–Perron and KPSS tests, with the right deterministic terms and lags
- ▣ Explain how a structural break can mimic a unit root
- ▣ Identify, estimate, check and forecast an  $ARIMA(p, d, q)$  model, and explain why its intervals grow with the horizon
- ▣ Use automatic ARIMA selection critically

## Reading and tools

- ▣ Textbook: [Huang and Petukhina \(2022\)](#), Ch. 4–5 (ARIMA models and non-stationarity)
  - ▶ companion, free online: [Hyndman and Athanasopoulos \(2021\)](#), Ch. 9 (ARIMA models)
- ▣ Theory: [Hamilton \(1994\)](#), Ch. 15–17 (trends, unit roots); [Box, Jenkins and Reinsel \(2008\)](#) for the ARIMA methodology
- ▣ Python Quantlets of this chapter: [Quantlets/Ch\\_03](#)
  - ▶ `adfuller`, `kpss`, `zivot_andrews` and ARIMA from `statsmodels`; the Phillips–Perron test written out in a few lines
- ▣ Lecture notebook: [open in Google Colab](#)
- ▣ Video course: [📺 Applied Time Series Analysis with Python](#)

## Four series that drift



- Romanian real GDP (Eurostat, seasonally adjusted) and HICP (Eurostat); EUR/RON, the BNR reference rate; BET and S&P 500 log prices, daily since 2000
- None of them has a constant mean: [Chapter 1](#) called them non-stationary

## Interpreting the four series

- GDP grows by 0.80% per quarter on average (about 3.2% per year), but it strays from the straight line by up to 18%
  - ▶ the 2009 recession: did GDP come back to the old line, or did it start a new one?
- Consumer prices:  $\times 7.2$  since 2000; EUR/RON: from 3.60 to 5.26 lei; BET: +426 log points, S&P 500: +166 log points (to 18 September 2026)
- Two models can produce such pictures
  - ▶ a fixed line plus stationary noise (**deterministic trend**)
  - ▶ a random walk with drift (**stochastic trend**)

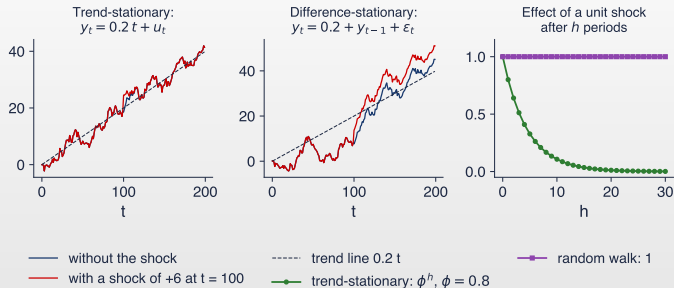
## Trend-stationary series

- **Definition:**  $y_t = \alpha + \beta t + u_t$ , with  $u_t$  a stationary process (for example a stationary AR(1), [Chapter 2](#))
  - ▶  $\alpha$ : the intercept;  $\beta$ : the slope, the average change per period;  $t$ : the time index
  - ▶  $y_t$  is **trend-stationary** (TS): stationary after subtracting the line  $\alpha + \beta t$
- Properties
  - ▶  $E[y_t] = \alpha + \beta t$ ;  $\text{Var}(y_t) = \text{Var}(u_t)$ , constant
  - ▶ a shock moves  $y_t$  away from the line only temporarily: with  $u_t = \phi u_{t-1} + \varepsilon_t$ , its effect after  $h$  periods is  $\phi^h$
  - ▶ long-run forecasts return to the line; their intervals have a bounded width
- Right treatment: estimate the trend by OLS (ordinary least squares) and model the residuals as ARMA

## Difference-stationary series

- **Definition:**  $y_t = \beta + y_{t-1} + u_t$ ,  $u_t$  stationary; by substitution,  $y_t = y_0 + \beta t + \sum_{i=1}^t u_i$ 
  - ▶  $\beta$ : the drift, the average change per period;  $y_0$ : the starting value
  - ▶  $y_t$  is **difference-stationary** (DS):  $\Delta y_t = \beta + u_t$  is stationary
  - ▶  $\sum_{i \leq t} u_i$  is the **stochastic trend**;  $\beta t$  is the drift
- Properties
  - ▶ the same mean  $y_0 + \beta t$  as a TS series, but  $\text{Var}(y_t)$  grows with  $t$  ([Chapter 1](#):  $t\sigma^2$  for a random walk)
  - ▶ every shock stays in the level forever: there is no line to return to
  - ▶ forecast intervals widen without limit as the horizon grows
- Right treatment: difference the series and model  $\Delta y_t$  as ARMA: the ARIMA model of this chapter

## Same drift, different memory



- Left and centre: the same shocks drive a TS series ( $u_t$  AR(1),  $\phi = 0.8$ ) and a random walk with drift 0.2; the red paths add one shock of +6 at  $t = 100$
- Right: the effect of a unit shock after  $h$  periods

## Interpreting the two trends

- TS series: the shock dies out; its effect is  $0.8^{10} = 0.107$  after 10 periods and 0.012 after 20
  - ▶ at  $t = 200$  the two red and blue paths coincide
- Random walk: the whole path shifts up by 6 and stays there
  - ▶ a recession in a DS economy is a permanent loss of output, not a dip below a trend
- By eye, the two series without the shock look alike: we need a formal test (Sections 3–4)

## Integrated processes

- **Definition:**  $y_t$  is **integrated of order  $d$** ,  $y_t \sim I(d)$ , if  $\Delta^d y_t$  is stationary and  $\Delta^{d-1} y_t$  is not
  - ▶  $I(0)$ : stationary (with an invertible ARMA representation, [Chapter 2](#));  $I(1)$ : stationary after one difference;  $I(2)$ : after two
  - ▶  $\Delta^2 y_t = \Delta y_t - \Delta y_{t-1} = y_t - 2y_{t-1} + y_{t-2}$
- Typical orders in economics and finance
  - ▶ log prices of stocks, exchange rates, log GDP:  $I(1)$ ; returns and growth rates:  $I(0)$
  - ▶ price levels in high-inflation periods: possibly  $I(2)$ , so that inflation itself is  $I(1)$
- Rules
  - ▶  $I(0) + I(1) = I(1)$ : the stochastic trend dominates; a TS series is  $I(0)$  after removing the trend, not after differencing

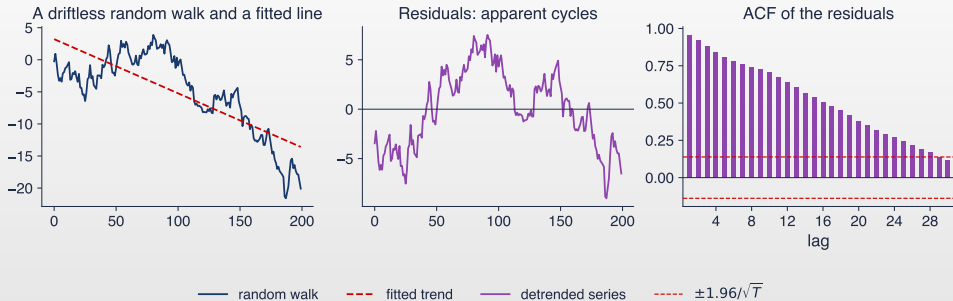
## The unit root

- Write an  $AR(p)$  with the lag operator  $L$  ([Chapter 1](#)):  $\phi(L)y_t = \varepsilon_t$ ,  $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$ 
  - ▶ [Chapter 2](#): stationary if all roots of  $\phi(z) = 0$  lie outside the unit circle,  $|z| > 1$
- **Unit root**: one root equals  $z = 1$ , so  $\phi(z) = (1 - z)\phi^*(z)$  with the roots of  $\phi^*$  outside the circle
  - ▶ then  $\phi^*(L)\Delta y_t = \varepsilon_t$ :  $\Delta y_t$  is a stationary  $AR(p-1)$ , and  $y_t \sim I(1)$
  - ▶  $AR(1)$ :  $\phi(z) = 1 - \phi z$  has the root  $1/\phi$ , a unit root when  $\phi = 1$ : the random walk
- **Example**:  $y_t = 1.5y_{t-1} - 0.5y_{t-2} + \varepsilon_t$ ;  $1 - 1.5z + 0.5z^2 = (1 - z)(1 - 0.5z)$ 
  - ▶ roots 1 and 2:  $\Delta y_t = 0.5\Delta y_{t-1} + \varepsilon_t$ , an  $AR(1)$  in differences

## Case study: Nelson and Plosser (1982)

- **Question:** are US macroeconomic series stationary around a trend, as the business-cycle models of the 1970s assumed?
  - ▶ Nelson and Plosser (1982): 14 long annual series (real GNP, employment, prices, wages, interest rates, stock prices), some since 1860
- **Method:** the Dickey–Fuller test of Section 3, with a constant and a trend
  - ▶ the unit root was rejected for one series only, the unemployment rate
- **Consequence:** shocks to output look permanent, not transitory
  - ▶ a large debate on the nature of recessions; and a warning against detrending by a straight line (next slide)
  - ▶ the answer was later challenged by structural breaks (Section 5)

## Detrending a random walk



□ Left: a driftless random walk ( $T = 200$ ) and the OLS line  $\hat{a} + \hat{b}t$ ; centre: the residuals; right: their ACF

## Interpreting the detrended random walk

- The line fits:  $R^2 = 0.61$ ,  $t = -17.7$  for the slope, although the true slope is 0
  - ▶ over 2000 random walks:  $|t| > 1.96$  in 90% of cases, median  $R^2 = 0.44$
- The residuals look like long cycles:  $\hat{\rho}(1) = 0.96$  and a slow decay
  - ▶ Nelson and Kang (1981): detrending a DS series creates **spurious cycles**; a business cycle may be an artefact of the method
- The reverse error (differencing a TS series) is over-differencing (Chapter 1 and Section 6)

## Recap: Deterministic and stochastic trends

- TS:  $y_t = \alpha + \beta t + u_t$ , shocks fade; DS:  $\Delta y_t = \beta + u_t$ , shocks are permanent
- $I(d)$ : stationary after  $d$  differences; a unit root in  $\phi(z)$  means  $I(1)$
- The wrong treatment creates artefacts: spurious cycles (detrending a DS series) or over-differencing (differencing a TS series)

## 1926: nonsense correlations

- Yule (1926) asked: *Why do we sometimes get nonsense-correlations between time-series?*
  - ▶ his example: in England and Wales, 1866–1911, the share of Church of England marriages and the mortality rate had a correlation of about 0.95
  - ▶ no causal link: both series simply declined over the period
- His explanation: for series that wander like sums of random terms, the sample correlation does not settle near 0
  - ▶ large values occur often by chance; the usual standard error does not apply
- Fifty years later, the same problem reappeared in econometric models estimated on levels

## 1974: Granger and Newbold

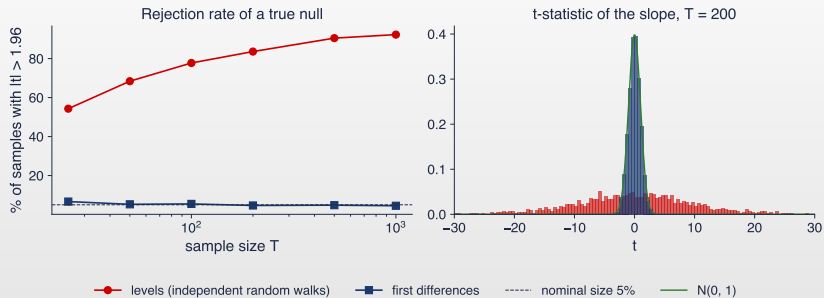


Clive Granger (1934–2009), Nobel Prize  
in Economics 2003

Photo: Olaf Storbeck (2008); CC BY-SA 2.0;  
[Wikimedia Commons](#)

- Granger and Newbold (1974): regress one random walk on another, **independent** random walk
  - ▶  $y_t = a + bx_t + e_t$  with  $y_t, x_t$  independent  $I(1)$  series
  - ▶ the  $t$ -test of  $b = 0$  rejects far too often;  $R^2$  is high; the Durbin–Watson statistic is close to 0
- Their rule of thumb: be suspicious when  $R^2 > DW$ 
  - ▶ DW (Durbin–Watson)  $\approx 2(1 - \hat{\rho}_1)$  of the residuals: near 0 means  $I(1)$ -like residuals
- Granger and Newbold were then at the University of Nottingham; Granger later received the Nobel Prize for cointegration ([Chapter 7](#))

## Spurious regression by simulation



- 2000 pairs of independent Gaussian random walks for each  $T$ ; OLS of  $y$  on a constant and  $x$ , in levels and in first differences
- Left: share of samples with  $|t| > 1.96$ ; right: the distribution of  $t$  for  $T = 200$

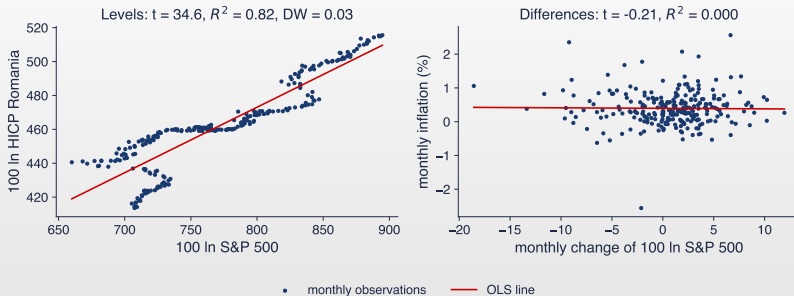
## Interpreting the simulation

- ▣ Levels: a true null  $b = 0$  is rejected in 68% of samples with  $T = 50$  and in 92% with  $T = 1000$ 
  - ▶ more data makes it worse: the median  $|t|$  grows from 3.2 ( $T = 50$ ) to 14.2 ( $T = 1000$ )
  - ▶ median  $R^2$  about 0.17 for every  $T$ ; median DW 0.08 at  $T = 200$
- ▣ Differences: 4.7% at  $T = 200$ , close to the nominal 5%
  - ▶ the  $t$ -test works again because  $\Delta y_t$  and  $\Delta x_t$  are stationary

## Why the $t$ -test fails

- Under  $b = 0$  the error  $e_t = y_t - a$  is itself a random walk: the OLS assumptions (stationary, uncorrelated errors) fail
  - ▶ the usual standard error is far too small
- Phillips (1986) derived the large-sample behaviour
  - ▶  $\hat{b}$  does not converge to 0: it converges to a random variable
  - ▶  $t_{\hat{b}}$  grows like  $\sqrt{T}$ , so the rejection rate tends to 100%
  - ▶  $R^2$  has a non-degenerate limit distribution;  $DW \rightarrow 0$
- Exception: if  $y_t - bx_t$  is stationary for some  $b$ , the series are **cointegrated** and the levels regression is meaningful Engle and Granger (1987) (Chapter 7)

## A spurious regression on real data



- Romanian HICP (Eurostat) against the S&P 500 (month-end close), 260 months since January 2005, both as 100 ln
- Left: levels; right: monthly changes

## Interpreting the HICP regression

- ▣ Levels: slope 0.39,  $t = 34.6$ ,  $R^2 = 0.82$ ,  $DW = 0.03$ 
  - ▶ “US stocks explain Romanian consumer prices”:  $R^2 > DW$ , a textbook spurious regression
- ▣ Differences:  $t = -0.21$  ( $p = 0.83$ ),  $R^2 = 0.0002$ 
  - ▶ monthly inflation and monthly stock returns are unrelated
- ▣ Both levels contain a trend; the regression only says that both went up

## Recap: Spurious regression

- Regressing one  $I(1)$  series on an unrelated  $I(1)$  series gives large  $t$ , high  $R^2$  and DW near 0
- More observations do not help:  $t$  grows like  $\sqrt{T}$
- Remedies: test the order of integration first; regress differences; or test for cointegration ([Chapter 7](#))

## Testing $\phi = 1$

- AR(1):  $y_t = \phi y_{t-1} + \varepsilon_t$ ; subtract  $y_{t-1}$ :  $\Delta y_t = \gamma y_{t-1} + \varepsilon_t$ , with  $\gamma = \phi - 1$ 
  - ▶  $H_0$ :  $\gamma = 0$  (unit root,  $I(1)$ );  $H_1$ :  $\gamma < 0$  (stationary); a one-sided, left-tail test
  - ▶ statistic:  $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$ , the usual OLS  $t$ -ratio; SE: the standard error; a very negative  $\tau$  speaks against the unit root
- Under  $H_0$ ,  $\tau$  is **not** Student or Normal [Dickey and Fuller \(1979\)](#)
  - ▶  $y_{t-1}$  is a random walk, so the regressor is not stationary;  $\hat{\phi}$  converges at rate  $T$ , not  $\sqrt{T}$  (**superconsistency**)
  - ▶ the limit of  $\tau$  is a functional of Brownian motion, skewed to the left [Hamilton \(1994\)](#), Ch. 17
  - ▶ critical values come from simulation: [Fuller \(1996\)](#); today from response surfaces [MacKinnon \(2010\)](#)

## Iowa State: where the test was born



Snedecor Hall, Iowa State University: the department of statistics  
and its Statistical Laboratory (1933)

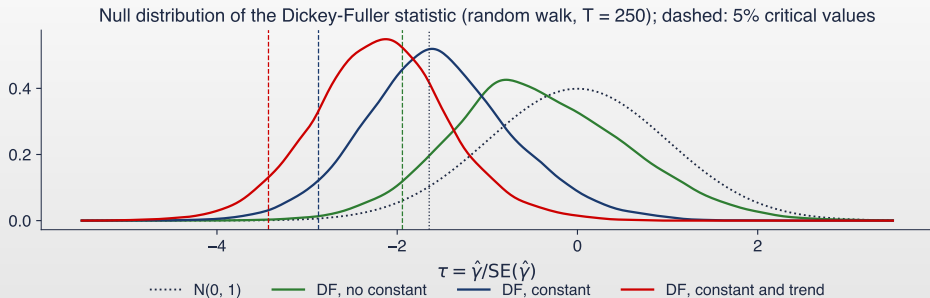
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- **Wayne A. Fuller**: statistician at Iowa State, author of *Introduction to Statistical Time Series* Fuller (1996)
- **David A. Dickey**: his doctoral student at Iowa State (1976), later at North Carolina State University
- **Dickey and Fuller (1979)**: the distribution of  $\hat{\phi}$  and  $\tau$  under a unit root; **Dickey and Fuller (1981)**: joint tests of the unit root and the deterministic terms
- **Said and Dickey (1984)**: the augmented test for ARMA errors (this section)

## Three Dickey–Fuller regressions

- **No constant:**  $\Delta y_t = \gamma y_{t-1} + \varepsilon_t$ 
  - ▶  $H_1$ : stationary with mean 0; rarely realistic
- **Constant:**  $\Delta y_t = c + \gamma y_{t-1} + \varepsilon_t$ 
  - ▶  $H_0$ : random walk (with drift if  $c \neq 0$ );  $H_1$ : stationary around a non-zero mean
  - ▶ for series without a trend: interest rates, inflation, exchange rates, returns
- **Constant and trend:**  $\Delta y_t = c + bt + \gamma y_{t-1} + \varepsilon_t$ 
  - ▶  $c$ : the constant;  $bt$ : a linear trend with slope  $b$
  - ▶  $H_0$ : random walk with drift;  $H_1$ : trend-stationary: exactly the TS against DS question of Section 1
  - ▶ for trending series: log GDP, log prices, log price indices
- Each case has its own distribution of  $\tau$  and its own critical values

## The Dickey-Fuller distributions



- 20 000 random walks with  $T = 250$ : the density of  $\tau$  in the three regressions, against  $N(0, 1)$ ; dashed lines: 5% critical values [MacKinnon \(2010\)](#)

## Interpreting the three distributions

- All three lie to the left of  $N(0, 1)$ ; the more deterministic terms, the further left
  - ▶ simulated 5% quantiles:  $-1.95$  (no constant),  $-2.86$  (constant) and  $-3.42$  (constant and trend); tabulated:  $-1.94$ ,  $-2.87$  and  $-3.43$
- With the Normal critical value  $-1.645$ , a true unit root is “rejected” in 46% of samples (constant) and 77% (constant and trend)
  - ▶ the wrong table turns random walks into “stationary” series

## Critical values of the Dickey–Fuller test

| $T$      | no constant |       |       | constant |       |       | constant and trend |       |       |
|----------|-------------|-------|-------|----------|-------|-------|--------------------|-------|-------|
|          | 1%          | 5%    | 10%   | 1%       | 5%    | 10%   | 1%                 | 5%    | 10%   |
| 50       | −2.61       | −1.95 | −1.61 | −3.57    | −2.92 | −2.60 | −4.15              | −3.50 | −3.18 |
| 100      | −2.59       | −1.94 | −1.61 | −3.50    | −2.89 | −2.58 | −4.05              | −3.46 | −3.15 |
| 250      | −2.57       | −1.94 | −1.62 | −3.46    | −2.87 | −2.57 | −4.00              | −3.43 | −3.14 |
| $\infty$ | −2.57       | −1.94 | −1.62 | −3.43    | −2.86 | −2.57 | −3.96              | −3.41 | −3.13 |

- Source: the response surfaces of [MacKinnon \(2010\)](#), as used by statsmodels; p-values from [MacKinnon \(1994\)](#)
- Decision: reject the unit root at 5% if  $\tau$  is **below** (more negative than) the 5% value
- The values depend only weakly on  $T$ ; they apply unchanged to the ADF test

## Worked example: Romanian real GDP

- $y_t = 100 \ln$  real GDP, 2000Q1–2026Q2; regression with constant and trend, no lags ( $T = 105$  differences)
  - ▶ OLS:  $\hat{\gamma} = -0.0961$ ,  $\text{SE}(\hat{\gamma}) = 0.0416$ , so  $\hat{\phi} = 1 + \hat{\gamma} = 0.904$
- $\tau = -0.0961/0.0416 = -2.31$ ; 5% critical value with constant and trend:  $-3.45$ 
  - ▶  $-2.31 > -3.45$ : do not reject the unit root ( $p = 0.43$ )
  - ▶ with the Normal value  $-1.645$  we would have rejected it: the wrong table gives the wrong answer
- Reading: “not rejected” is not “proved”
  - ▶ if  $\phi = 0.904$  were true, a shock would halve in  $\ln 0.5 / \ln 0.904 \approx 6.9$  quarters; with 106 observations the test can hardly tell this from  $\phi = 1$

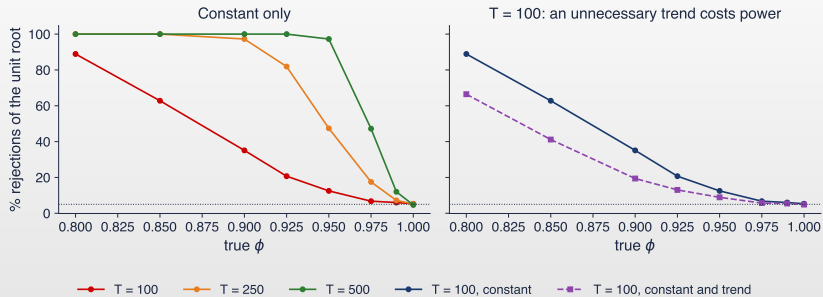
## The augmented Dickey–Fuller test

- If  $\Delta y_t$  is autocorrelated,  $\varepsilon_t$  in the DF regression is not white noise and the size of the test is wrong
  - ▶ the remedy: add lagged differences
- **ADF** (augmented Dickey–Fuller) regression:
  - ▶  $\Delta y_t = c + bt + \gamma y_{t-1} + \sum_{j=1}^k \delta_j \Delta y_{t-j} + \varepsilon_t$ ; same  $\tau$ , same critical values
  - ▶  $\delta_j$ : the coefficients of the past differences;  $k$ : the number of lagged differences
  - ▶ an  $\text{AR}(p)$  in levels becomes exactly this regression with  $k = p - 1$  and  $\gamma = -\phi(1)$
- **Said and Dickey (1984)**: with ARMA errors the test stays valid if  $k$  grows slowly with  $T$ 
  - ▶ the lags approximate the MA part by a long AR
- The ADF test is the default unit-root test in software: `adfuller` in `statsmodels`

## Choosing the number of lags $k$

- Maximum:  $k_{\max} = 12 (T/100)^{1/4}$  [Schwert \(1989\)](#); for  $T = 105$ ,  $k_{\max} = 13$ 
  - ▶ then choose  $k \leq k_{\max}$  by an information criterion, AIC or BIC ([Chapter 2](#)), on the same sample for every  $k$
- Too few lags: autocorrelated errors, wrong size (with a negative MA part, far too many rejections)
  - ▶ too many lags: lower power
- [Ng and Perron \(2001\)](#): a modified AIC (MAIC) gives a better size; GLS detrending [Elliott, Rothenberg and Stock \(1996\)](#) gives more power
  - ▶ MAIC: modified Akaike information criterion; GLS: generalised least squares; both appear in the DF-GLS test
- Check the result: Ljung–Box on the ADF residuals [Ljung and Box \(1978\)](#); report  $k$  together with  $\tau$

## Size and power of the Dickey–Fuller test



□ 2000 AR(1) paths  $y_t = \phi y_{t-1} + \varepsilon_t$  for each  $\phi$ ; share of rejections at 5%; at  $\phi = 1$  this is the size

 TSA\_ch3\_dickey\_fuller

## Interpreting the power curves

- ▣ Size: 5%, 5% and 5% at  $\phi = 1$ : the test holds its 5% level
- ▣ Power at  $\phi = 0.95$ : 12% with  $T = 100$ , 47% with  $T = 250$ , 97% with  $T = 500$ 
  - ▶ a quarterly series of 25 years ( $T = 100$ ) almost never distinguishes  $\phi = 0.95$  from a unit root
  - ▶ what matters is the time span, not the frequency: daily data over two years do not help
- ▣ An unnecessary trend costs power: at  $T = 100$  and  $\phi = 0.9$ , 35% with a constant, 19% with constant and trend

## Choosing the deterministic terms

- The alternative must be a plausible description of the data
  - ▶ trending series (log GDP, log prices): constant and trend
  - ▶ series that fluctuate around a level (interest rates, inflation, exchange rates, returns): constant
- Omitting a needed trend: a TS series looks like a unit root; the test almost never rejects
  - ▶ adding an unneeded trend: lower power (previous chart)
- Look at the plot first, then choose; [Dickey and Fuller \(1981\)](#) give joint  $F$ -type tests of the unit root and the trend

## How many differences: $I(1)$ or $I(2)$ ?

- Dickey and Pantula (1987): test from the highest plausible order downwards
  - ▶ step 1: ADF on  $\Delta y_t$ ; if it does not reject,  $y_t$  may be  $I(2)$
  - ▶ step 2: if it rejects, ADF on  $y_t$ ; reject:  $I(0)$ ; do not reject:  $I(1)$
- Why downwards: if  $y_t \sim I(2)$ , the test on  $y_t$  assumes at most one unit root and is not valid
- Example: Romanian consumer prices since 2005 (the table in Section 4)
  - ▶ In  $P$ : unit root; inflation  $\pi$ : ADF  $p = 0.32$ , KPSS 0.404; change of inflation: clearly stationary
  - ▶ so  $\ln P$  is  $I(1)$  or  $I(2)$ , depending on how persistent inflation is: Section 7 returns to this

## Recap: The Dickey–Fuller test

- ▣  $\Delta y_t = c + bt + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$ ;  $H_0: \gamma = 0$ ;  $\tau$  against Dickey–Fuller critical values
- ▣ Deterministic terms follow the plot; lags by AIC or BIC up to  $12(T/100)^{1/4}$
- ▣ Low power near  $\phi = 1$ : not rejecting is weak evidence for a unit root
- ▣ Order of integration: test from the top ( $\Delta y_t$  first)

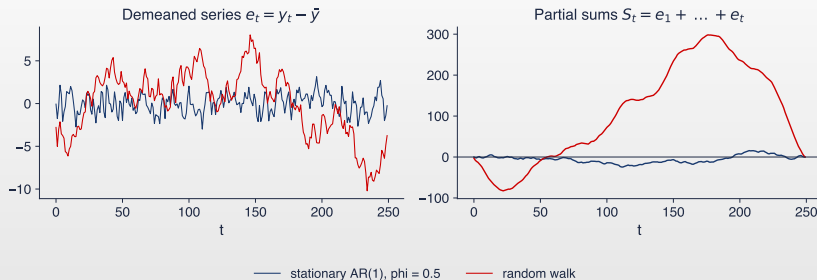
## The Phillips–Perron test

- ▣ **Phillips and Perron (1988)**: keep the simple DF regression (no lagged differences) and correct  $\tau$  instead
  - ▶  $Z_\tau = \sqrt{\hat{\gamma}_0 / \hat{\lambda}^2} \tau - \frac{(\hat{\lambda}^2 - \hat{\gamma}_0) T \text{SE}(\hat{\gamma})}{2 \hat{\lambda} s}$
  - ▶  $\hat{\gamma}_0$ : variance of the residuals;  $s^2$ : their OLS variance;  $\hat{\lambda}^2$ : their **long-run variance**
- ▣ Long-run variance (Newey–West, Bartlett weights):  $\hat{\lambda}^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^L (1 - \frac{j}{L+1}) \hat{\gamma}_j$ 
  - ▶  $\hat{\gamma}_j$ : autocovariances of the residuals;  $L$ : the number of autocovariances included (the bandwidth)
  - ▶ if the residuals are white noise,  $\hat{\lambda}^2 = \hat{\gamma}_0$  and  $Z_\tau = \tau$ : the correction only matters when they are autocorrelated
- ▣ Same  $H_0$  and the same critical values as ADF; robust to heteroskedasticity; no lag choice, but a bandwidth  $L$ 
  - ▶ weak point: large size distortions with a negative MA component **Schwert (1989)**

## The KPSS test: stationarity as the null

- ▣ Kwiatkowski et al. (1992) (KPSS: Kwiatkowski–Phillips–Schmidt–Shin):  $y_t = \xi t + r_t + \varepsilon_t$ ,  $r_t = r_{t-1} + u_t$ ,  $u_t \sim \text{WN}(0, \sigma_u^2)$ 
  - ▶  $\xi t$ : a deterministic trend;  $r_t$ : a random walk, the moving level;  $\varepsilon_t$ : stationary noise
  - ▶  $H_0$ :  $\sigma_u^2 = 0$  (the random walk is a constant:  $y_t$  stationary around a level or a trend);  $H_1$ :  $\sigma_u^2 > 0$ , a unit root
- ▣ Statistic: regress  $y_t$  on a constant (and a trend); residuals  $e_t$ ; partial sums  $S_t = e_1 + \dots + e_t$ 
  - ▶  $\eta = \frac{1}{T^2 \hat{\lambda}^2} \sum_{t=1}^T S_t^2$ , with  $\hat{\lambda}^2$  the long-run variance of  $e_t$
  - ▶ reject stationarity for **large**  $\eta$ : 5% critical values 0.463 (level) and 0.146 (trend); 1%: 0.739 and 0.216
- ▣ A test of stationarity, not of a unit root: it reverses the burden of proof

## KPSS: why partial sums



- A stationary AR(1) ( $\phi = 0.5$ ) and a random walk,  $T = 250$ : the demeaned series and their partial sums  $S_t$
- KPSS (level): 0.16 for the AR(1), 0.73 for the random walk; 5% critical value 0.463

## Interpreting the partial sums

- Stationary series: deviations cancel out,  $S_t$  stays close to 0
  - ▶  $\sum S_t^2$  grows like  $T^2$ , so  $\eta$  stays bounded
- Random walk: long runs on one side of the mean,  $S_t$  makes large excursions
  - ▶  $\sum S_t^2$  grows like  $T^4$ :  $\eta$  grows with  $T$  and the test rejects
- The same idea as the CUSUM charts of quality control: cumulated deviations reveal a drifting level

## ADF and KPSS together

|                            | KPSS does not reject                 | KPSS rejects                            |
|----------------------------|--------------------------------------|-----------------------------------------|
| <b>ADF rejects</b>         | $I(0)$ : both agree                  | conflict: break, long memory, outliers? |
| <b>ADF does not reject</b> | inconclusive: the data cannot decide | $I(1)$ : both agree                     |

- The two tests have opposite null hypotheses: agreement is stronger evidence than either test alone
- Inconclusive is common with short samples: low power on both sides
  - ▶ then decide by the purpose: for forecasting,  $d = 1$  is the cautious choice (wider intervals)
- Conflict: look for breaks (Section 5) or fractional integration ([Chapter 8](#))

## Unit-root tests on real series

| Series                            | det. | $T$   | ADF (p)          | $k$ | PP     | KPSS   | ADF+KPSS     |
|-----------------------------------|------|-------|------------------|-----|--------|--------|--------------|
| Romania, real GDP, 100 ln $Y$     | c, t | 106   | -2.31 (0.428)    | 0   | -2.19  | 0.198  | $I(1)$       |
| its growth $\Delta$               | c    | 105   | -10.77 (< 0.001) | 0   | -10.88 | 0.263  | $I(0)$       |
| Romania, HICP, 100 ln $P$         | c, t | 260   | -0.96 (0.949)    | 6   | -0.88  | 0.314  | $I(1)$       |
| Romania, 12-month inflation $\pi$ | c    | 260   | -1.94 (0.315)    | 15  | -2.31  | 0.404  | inconclusive |
| its change $\Delta\pi$            | c    | 259   | -4.83 (< 0.001)  | 14  | -11.31 | 0.105  | $I(0)$       |
| US, real GDP, 100 ln $Y$          | c, t | 318   | -1.54 (0.816)    | 1   | -1.20  | 0.581  | $I(1)$       |
| its growth $\Delta$               | c    | 317   | -15.51 (< 0.001) | 0   | -15.44 | 0.456  | $I(0)$       |
| EUR/RON, 100 ln $S$               | c    | 5 353 | -1.22 (0.664)    | 7   | -1.15  | 10.255 | $I(1)$       |
| daily returns                     | c    | 5 352 | -27.42 (< 0.001) | 6   | -60.82 | 0.055  | $I(0)$       |
| BET, 100 ln $P$                   | c, t | 6 671 | -1.90 (0.653)    | 19  | -2.35  | 1.143  | $I(1)$       |
| daily returns                     | c    | 6 670 | -16.50 (< 0.001) | 18  | -74.55 | 0.360  | $I(0)$       |
| S&P 500, 100 ln $P$               | c, t | 6 715 | -2.08 (0.557)    | 34  | -2.36  | 2.281  | $I(1)$       |
| daily returns                     | c    | 6 714 | -14.26 (< 0.001) | 35  | -91.41 | 0.449  | $I(0)$       |
| Nile, yearly flow                 | c    | 100   | -4.05 (0.001)    | 1   | -6.38  | 0.869  | conflict     |

- det.: deterministic terms (c: constant; t: trend);  $k$ : ADF lags (AIC); 5% critical values: ADF and PP about -2.86 (c), -3.41 (c, t); KPSS 0.463 (c), 0.146 (c, t)

## Interpreting the unit-root table

- Log prices, the exchange rate, log GDP and log HICP:  $I(1)$ ; their differences:  $I(0)$ 
  - ▶ Romanian GDP: ADF  $p = 0.43$  and KPSS  $0.198 > 0.146$ : a unit root, as [Nelson and Plosser \(1982\)](#) found for the US
- Romanian 12-month inflation: neither test rejects (KPSS  $0.404 < 0.463$ ): inconclusive
  - ▶ a very persistent series, possibly with breaks; Sections 5 and 7
- Nile: both reject: a conflict
  - ▶ [Chapter 1](#) showed a level shift in 1898: the next section explains the conflict
- Daily returns: PP gives far more negative values than ADF (BET:  $-74.6$  against  $-16.5$ ), but both reject; the verdict is the same

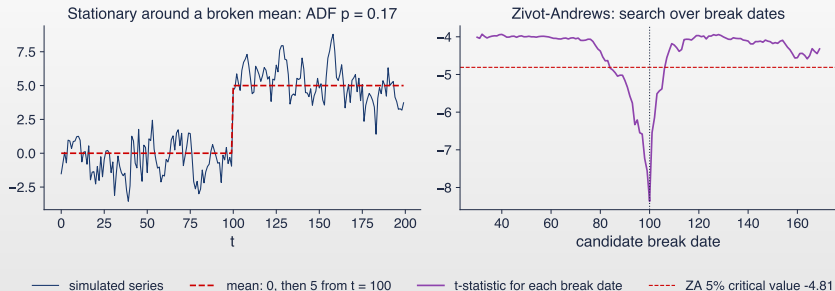
## Recap: Phillips–Perron and KPSS

- PP: DF regression plus a long-run-variance correction; same  $H_0$  and critical values as ADF
- KPSS:  $H_0$  stationarity; large partial sums reject it
- Read ADF and KPSS together: agree, inconclusive, or conflict

## Perron (1989): a break can look like a unit root

- Perron (1989): a stationary series around a mean or a trend that **shifts once** looks persistent
  - ▶ the DF regression explains the shift by  $\hat{\phi}$  close to 1; the test loses power
- He re-examined Nelson and Plosser with two known breaks: the 1929 crash (level) and the 1973 oil shock (slope)
  - ▶ with dummy variables for the break, the unit root was rejected for most series
  - ▶ reading: most shocks are transitory; only a few rare events are permanent
- The critique: the break date was chosen after looking at the data

## A level shift misleads the ADF test



- Left: AR(1) noise ( $\phi = 0.6$ ) around a mean that jumps from 0 to 5 at  $t = 100$ ; right: the  $t$ -statistic on  $y_{t-1}$  for every candidate break date, with a level dummy

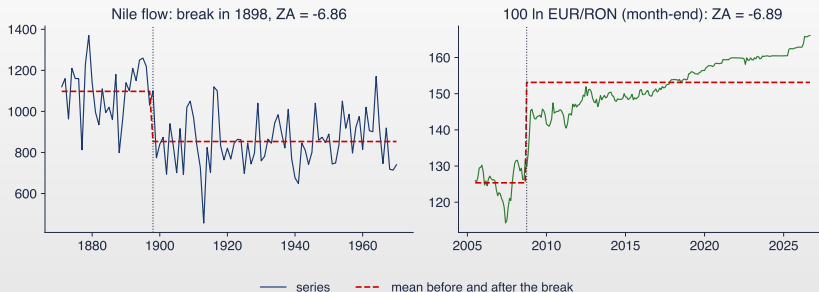
## Interpreting the simulated break

- ADF with a constant:  $p = 0.17$ : the stationary series “has a unit root”
  - ▶ over 2000 such series, the DF test rejects in 72% of cases; without the break: 100%
- With the break modelled, the statistic falls to  $-7.47$ , far below the 5% value  $-4.81$ 
  - ▶ the minimum is reached at the true date  $t = 100$

## Zivot and Andrews (1992): an unknown break date

- Zivot and Andrews (1992): let the data choose the break date  $T_B$ 
  - ▶ for each  $T_B$  in the central 70% of the sample, run the ADF regression with a break dummy:  
 $\Delta y_t = c + \theta DU_t + bt + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$ ,  $DU_t = 1$  for  $t > T_B$
  - ▶  $DU_t$ : a dummy variable, 0 before and 1 after the break;  $\theta$ : the size of the level shift
  - ▶ the statistic is the **minimum**  $t$ -ratio of  $\gamma$  over all dates
- Three versions: break in the level, in the slope, in both
  - ▶ critical values more negative than ADF: about  $-4.81$  (level) and  $-5.07$  (both) at 5%
- $H_0$ : unit root without a break;  $H_1$ : stationary with one break
  - ▶ several breaks: Bai and Perron (1998); a rejection does not prove that the break is the right story

## Zivot–Andrews on two real series



- Left: the Nile flow (statsmodels data set, 1871–1970); right: 100 ln EUR/RON, month-end BNR reference rate
- Dotted line: the break date chosen by the test (break in the level)

## Interpreting the two breaks

- Nile:  $ZA = -6.86$ , break in 1898, the mean falls from 1098 to 853
  - ▶ the date matches the history: the first Aswan dam (built 1898–1902) and a drier climate; the ADF–KPSS conflict is explained
- EUR/RON: ADF  $p = 0.55$ , but  $ZA = -6.89$ , break in October 2008: the 2008 depreciation (mean 3.50 lei before, 4.62 after)
  - ▶ a managed exchange rate: long calm periods and rare adjustments, not a pure random walk
  - ▶ but after 2008 the rate still drifts up: “stationary around one break” is also a simplification

## Recap: Structural breaks

- ▣ A one-time shift in the mean or trend biases unit-root tests towards non-rejection
- ▣ Zivot–Andrews searches the break date; its critical values are more negative
- ▣ Unit root or break: check the dates against known events

## The ARIMA model

- **Definition:**  $y_t \sim \text{ARIMA}(p, d, q)$  (autoregressive integrated moving average) if  $\Delta^d y_t$  is a stationary, invertible ARMA( $p, q$ ):
  - ▶  $\phi(L)(1-L)^d y_t = c + \theta(L)\varepsilon_t$ ,  $\varepsilon_t \sim \text{WN}(0, \sigma^2)$
  - ▶  $p$ : AR order;  $d$ : number of differences (unit roots);  $q$ : MA order;  $\phi(L)$ ,  $\theta(L)$  as in [Chapter 2](#)
- The AR polynomial of the levels is  $\phi(z)(1-z)^d$ :  $d$  roots exactly on the unit circle
  - ▶ “integrated”:  $y_t$  is a  $d$ -fold cumulative sum of an ARMA process
- [Box, Jenkins and Reinsel \(2008\)](#) made the model popular; in practice  $d \leq 2$ , and  $d = 1$  most often

## Special cases and the constant

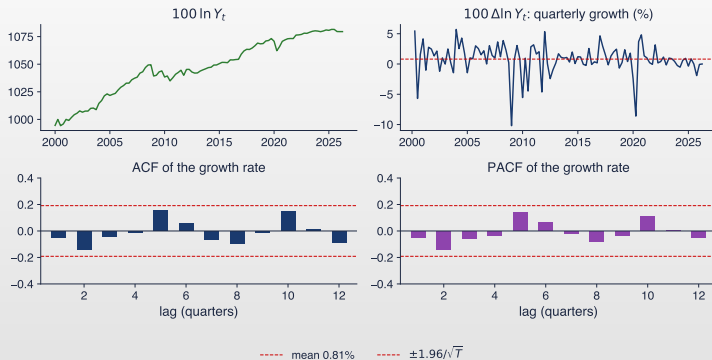
| Model        | Equation and meaning                                                                                                                            |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ARIMA(0,1,0) | $\Delta y_t = c + \varepsilon_t$ : random walk (with drift $c$ )                                                                                |
| ARIMA(1,1,0) | $\Delta y_t = c + \phi \Delta y_{t-1} + \varepsilon_t$ : growth with momentum                                                                   |
| ARIMA(0,1,1) | $\Delta y_t = \varepsilon_t + \theta \varepsilon_{t-1}$ : simple exponential smoothing with $\alpha = 1 + \theta$ ( <a href="#">Chapter 0</a> ) |
| ARIMA(0,2,2) | $\Delta^2 y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$ : the model behind Holt's linear method                |

- The constant  $c$  changes the long-run forecast [Hyndman and Athanasopoulos \(2021\)](#), Ch. 9.7
  - ▶  $d = 0$ : forecasts go to the mean;  $d = 1$ :  $c \neq 0$  gives a straight line (drift),  $c = 0$  a flat line
  - ▶  $d = 2$ :  $c \neq 0$  gives a quadratic trend, rarely sensible; use  $c = 0$

## The Box–Jenkins cycle with unit roots

- ▣ 1. **Transform**: logarithm or Box–Cox if the variance grows with the level ([Chapter 1](#))
- ▣ 2. **Choose**  $d$ : plot, ACF, ADF and KPSS; test  $\Delta y_t$  first; stop at the first stationary difference
- ▣ 3. **Choose**  $p, q$ : ACF and PACF of  $\Delta^d y_t$  ([Chapter 2](#)), then compare candidates by AICc on the same  $d$
- ▣ 4. **Estimate** by maximum likelihood; decide on the constant
- ▣ 5. **Check** the residuals: ACF, Ljung–Box with  $m - p - q$  degrees of freedom, outliers
- ▣ 6. **Forecast** the levels with intervals; go back to step 2 or 3 if the checks fail

## Romanian GDP: identification



- Real GDP, seasonally adjusted (Eurostat), 2000Q1–2026Q2,  $T = 106$ ; top:  $100 \ln Y_t$  and the quarterly growth; bottom: ACF and PACF of the growth

## Interpreting the identification plots

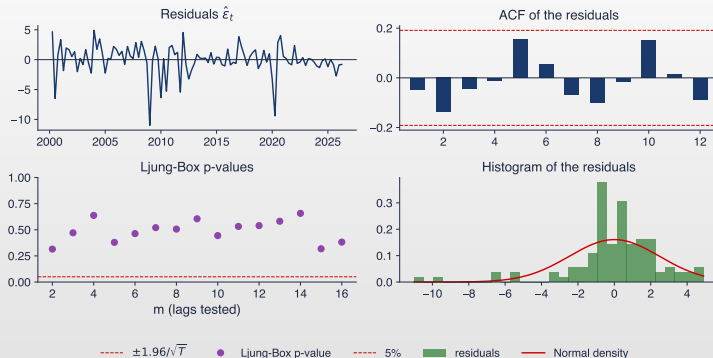
- ▣  $d = 1$ : the level is  $I(1)$  (Section 4); the growth rate is stationary, mean 0.81% per quarter (about 3.2% per year)
  - ▶ two large falls:  $-10.2\%$  in 2009Q1 (global financial crisis) and  $-8.6\%$  in 2020Q2 (pandemic)
- ▣  $p$  and  $q$ : no ACF or PACF value outside  $\pm 0.19$ ;  $\hat{\rho}(1) = -0.05$ ,  $\hat{\rho}(2) = -0.14$ 
  - ▶ Ljung–Box  $Q^*(8) = 7.3$  ( $p = 0.51$ ): the growth rate looks like white noise around its mean
- ▣ The plots suggest ARIMA(0,1,0) with drift; the next slide checks the alternatives

## Model choice by AICc

|         | $q = 0$ | $q = 1$ | $q = 2$            |
|---------|---------|---------|--------------------|
| $p = 0$ | 491.9   | 493.7   | 493.6              |
| $p = 1$ | 493.8   | 494.9   | 495.8              |
| $p = 2$ | 493.7   | 495.7   | 495.4 <sup>†</sup> |

- ARIMA( $p, 1, q$ ) with drift for  $100 \ln Y_t$ , estimated by maximum likelihood; AICc: AIC with the small-sample correction ([Chapter 2](#)); <sup>†</sup>: the optimiser did not converge
- Lowest AICc: ARIMA(0,1,0); the next model is 1.7 points worse
  - differences below 2 points are not meaningful: keep the simplest model
- Estimate: drift  $\hat{c} = 0.81$  (SE 0.30,  $t = 2.72$ ),  $\hat{\sigma} = 2.47$ :  $100 \Delta \ln Y_t = 0.81 + \hat{\varepsilon}_t$

## Romanian GDP: residual checks



- ☐ Residuals of ARIMA(0,1,0) with drift: time plot, ACF, Ljung-Box p-values for  $m = 2, \dots, 16$ , histogram with a Normal density

## Interpreting the residual checks

- ▣ No autocorrelation left:  $Q^*(8) = 7.3$  ( $p = 0.51$ ),  $Q^*(12)$   $p = 0.54$ ; all p-values above 5%
- ▣ Not Normal: kurtosis 7.9, Jarque–Bera 145; the largest residual is  $-11.0$  in 2009Q1
  - ▶ two crises dominate the tails; the Normal intervals of the next slides are too narrow in a crisis and slightly too wide in calm times
  - ▶ options: dummy variables for 2009 and 2020, or a bootstrap of the residuals for the intervals

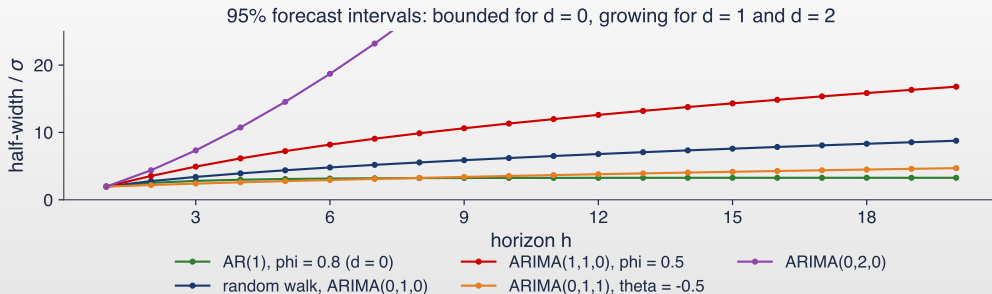
## Forecasting with ARIMA

- ▣ **Point forecasts:** write the model for the levels and iterate it, with  $\hat{\varepsilon}_{T+j} = 0$  for  $j \geq 1$ 
  - ▶ ARIMA(1,1,0):  $y_t = (1 + \phi)y_{t-1} - \phi y_{t-2} + \varepsilon_t$
- ▣ **Forecast error:**  $y_{T+h} - \hat{y}_{T+h} = \sum_{j=0}^{h-1} \psi_j \varepsilon_{T+h-j}$ , with  $\psi(z) = \theta(z)/[\phi(z)(1-z)^d]$ 
  - ▶ variance  $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ ; 95% interval  $\hat{y}_{T+h} \pm 1.96 \sigma \sqrt{\sum_{j=0}^{h-1} \psi_j^2}$
  - ▶ for  $d \geq 1$  the  $\psi_j$  do not decay to 0: the variance grows without bound
- ▣ Two closed forms
  - ▶ random walk:  $\psi_j = 1$ , variance  $h\sigma^2$ : the interval grows like  $\sqrt{h}$
  - ▶ ARIMA(0,1,1):  $\psi_j = 1 + \theta$  for  $j \geq 1$ , variance  $\sigma^2[1 + (h-1)(1+\theta)^2]$

## Worked example: ARIMA(1,1,0) forecasts

- ▣  $\Delta y_t = 0.6 \Delta y_{t-1} + \varepsilon_t$ ,  $\sigma = 2$ ; last data:  $y_T = 108$ ,  $\Delta y_T = 5$
- ▣ **Point forecasts:** forecast the change, then add it
  - ▶  $\widehat{\Delta y}_{T+1} = 0.6 \cdot 5 = 3$ ,  $\hat{y}_{T+1} = 111.00$ ;  $\widehat{\Delta y}_{T+2} = 1.8$ ,  $\hat{y}_{T+2} = 112.80$ ;  $\hat{y}_{T+3} = 113.88$
- ▣  $\psi$  **weights:**  $\psi_0 = 1$ ,  $\psi_1 = 1 + \phi = 1.60$ ,  $\psi_2 = 1 + \phi + \phi^2 = 1.96$ 
  - ▶ variances:  $4$ ,  $4(1 + 1.60^2) = 14.24$ ,  $4(1 + 1.60^2 + 1.96^2) = 29.61$
  - ▶ 95% half-widths: 3.92, 7.40, 10.66: the interval almost triples in three steps
- ▣ The momentum  $\phi = 0.6$  makes the uncertainty grow faster than for a random walk ( $\sqrt{3} = 1.73$  times in three steps)

## How fast the intervals grow



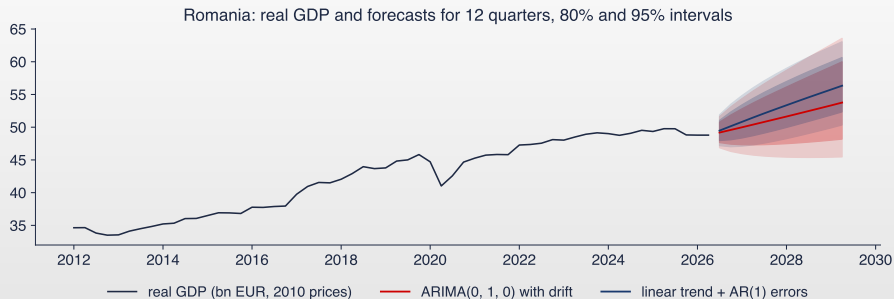
□ Half-width of the 95% forecast interval, in units of  $\sigma$ , for horizons 1 to 20, from the  $\psi$  weights of five models

 TSA\_ch3\_forecast\_intervals

## Interpreting the interval widths

- ▣  $d = 0$ , AR(1) with  $\phi = 0.8$ : the width levels off at  $1.96\sigma / \sqrt{1 - \phi^2}$  ( $3.27\sigma$  at  $h = 20$ )
- ▣  $d = 1$ : growth like  $\sqrt{h}$ ; at  $h = 20$ :  $8.77\sigma$  (random walk),  $16.78\sigma$  (ARIMA(1,1,0)),  $4.70\sigma$  (ARIMA(0,1,1),  $\theta = -0.5$ )
  - ▶ a negative MA part offsets part of each shock: narrower intervals
- ▣  $d = 2$ : growth like  $h^{3/2}$  ( $105.00\sigma$  at  $h = 20$ ): over-differencing makes long-run forecasts useless

## Romanian GDP: two forecasts



- ARIMA(0,1,0) with drift (red) and a linear trend with AR(1) errors (blue), both estimated on 2000Q1–2026Q2; forecasts to 2029Q2, transformed back to bn EUR

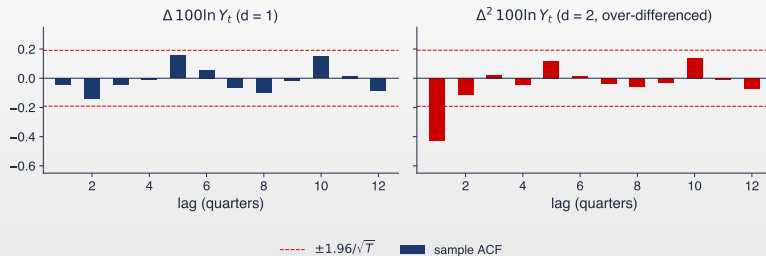
## Interpreting the GDP forecasts

- ▣ The random walk with drift grows by 0.81% per quarter from the last value (48.8 bn EUR in 2026Q2): 53.8 bn after 12 quarters
  - ▶ 95% half-width (in % of GDP): 4.8 after 1 quarter, 9.7 after 4, 16.8 after 12, i.e.  $\times 3.46 = \sqrt{12}$
- ▣ The TS model pulls GDP back towards its old line ( $\hat{\phi} = 0.92$ ): 56.3 bn after 12 quarters, half-width 11.2%
  - ▶ more optimistic and more confident, because it assumes the recent slowdown is temporary
- ▣ The tests could not settle TS against DS for this sample: the choice of  $d$  is a forecasting decision with consequences

## Over-differencing

- Differencing an  $I(0)$  series creates an MA unit root:  $\Delta\varepsilon_t = \varepsilon_t - \varepsilon_{t-1}$ ,  $\theta = -1$  (Chapter 1)
  - ▶ the result is not invertible: no  $AR(\infty)$  representation, poor estimation Plosser and Schwert (1977)
- Symptoms
  - ▶  $\hat{\rho}(1)$  of the differenced series close to  $-0.5$
  - ▶ the variance increases after differencing instead of decreasing
  - ▶ an estimated MA coefficient close to  $-1$
- Rule: difference only as long as the variance goes down and the tests ask for it

## Over-differencing Romanian GDP



□ ACF of the quarterly growth  $\Delta 100 \ln Y_t (d = 1)$  and of its difference  $\Delta^2 100 \ln Y_t (d = 2)$

 TSA\_ch3\_arima\_gdp

## Interpreting over-differencing

- ▣  $d = 1$ :  $\hat{\rho}(1) = -0.05$ , variance 6.16;  $d = 2$ :  $\hat{\rho}(1) = -0.43$ , variance 12.80: all three symptoms
- ▣ An MA(1) fitted to  $\Delta^2 100 \ln Y_t$  gives  $\hat{\theta} = -0.987$  (SE 0.080): practically  $-1$ , the extra difference is cancelled by the MA part
- ▣ Correct choice:  $d = 1$

## Recap: ARIMA models

- ▣ ARIMA( $p, d, q$ ): an ARMA( $p, q$ ) for  $\Delta^d y_t$ ; the constant decides the long-run slope
- ▣ Romanian GDP since 2000: ARIMA(0,1,0) with drift, 0.81% per quarter; residuals uncorrelated but heavy-tailed
- ▣ Forecast variance  $\sigma^2 \sum_{j < h} \psi_j^2$ : bounded for  $d = 0$ , growing for  $d \geq 1$
- ▣ Over-differencing:  $\hat{\rho}(1) \approx -0.5$ , larger variance, MA coefficient near  $-1$

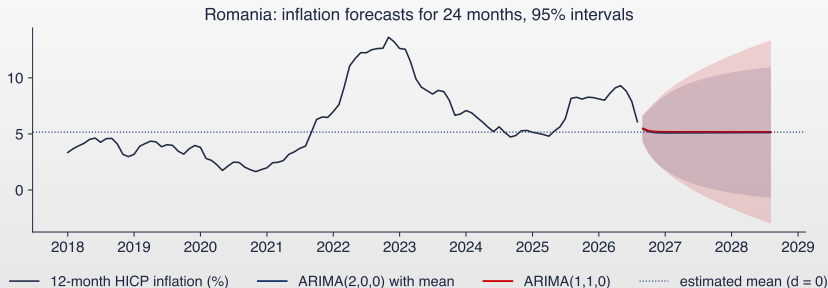
## Automatic ARIMA

- ▣ Hyndman and Khandakar (2008) (`auto.arima` in R; `pmdarima` and `statsforecast` in Python):
  - ▶  $d$ : repeated KPSS tests at 5%: difference while KPSS rejects
  - ▶  $p, q$ : a stepwise search from a few starting models, by AICc, on the chosen  $d$ ; a constant or drift if  $d \leq 1$
  - ▶ rejects models with roots too close to the unit circle and fits that fail
- ▣ Useful for many series at once and as a starting point
  - ▶ evaluated by its authors on the 3 003 series of the M3 forecasting competition, it was competitive with the best methods there
- ▣ But each step embeds a choice that you must be able to defend (next slide)

## Automatic ARIMA: checks

- ▣ **AICc is not comparable across  $d$ :** with  $d = 1$  the likelihood is computed for  $\Delta y_t$ , a different series
  - ▶ choose  $d$  by tests and judgement, then compare  $p$ ,  $q$
- ▣ **Near-cancelling roots:** Romanian GDP from 1995Q1: the lowest AICc, 607.5 (ARIMA(0,1,0): 611.1), belongs to ARIMA(2,1,2), whose fit did not converge
  - ▶ MA roots of modulus 1.0002, AR roots 1.014: AR and MA almost cancel;  $\hat{\theta}_2 = 1.000$  with SE 6.96
  - ▶ a model fitted to the noise of the 1990s data, not a description of the economy
- ▣ **Always:** plot the data, check the convergence and the roots, run the residual tests, compare with a simple benchmark (random walk, [Chapter 0](#))

## Romanian inflation: $d = 0$ or $d = 1$ ?



- 12-month HICP inflation (Eurostat), January 2005–August 2026; best model by AICc for  $d = 0$  (with a mean) and for  $d = 1$ ,  $p \leq 3$ ,  $q \leq 2$ ; forecasts for 24 months

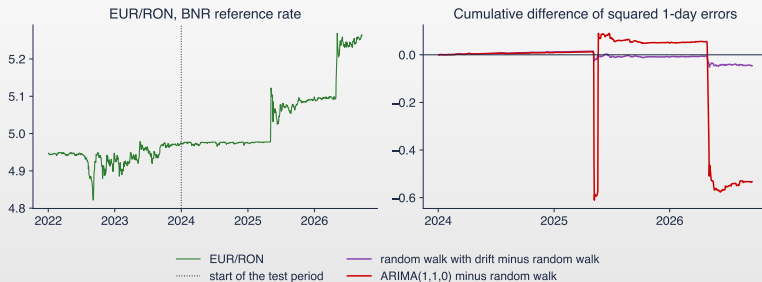
## Interpreting the two inflation models

- $d = 0$ : ARIMA(2,0,0),  $\hat{\phi}_1 = 1.32$ ,  $\hat{\phi}_2 = -0.34$ , sum 0.976: stationary, but barely; mean 5.2%
  - ▶  $d = 1$ : ARIMA(1,1,0),  $\hat{\phi} = 0.33$ : almost the same short-run dynamics
- After 24 months: 5.1% in  $[-0.6, 10.9]$  against 5.2% in  $[-2.9, 13.3]$ 
  - ▶ similar points, different intervals:  $d$  decides how much uncertainty we report
- KPSS (0.404) keeps  $d = 0$ , as automatic ARIMA would; ADF ( $p = 0.32$ ) suggests  $d = 1$ 
  - ▶ both models leave autocorrelated residuals: 12-month rates overlap and inherit the season; [Chapter 4](#) models monthly inflation

## Case study: Meese and Rogoff (1983)

- ▣ **Question:** can exchange-rate models forecast better than a random walk?
  - ▶ **Meese and Rogoff (1983):** monetary models of the 1970s, time series models (ARIMA, VAR) and the random walk; dollar rates, horizons of 1 to 12 months
- ▣ **Finding:** out of sample, none beat the random walk in RMSE (root mean squared error)
  - ▶ even with the actual future values of the fundamentals plugged in
- ▣ **Legacy:** the “Meese–Rogoff puzzle” and the random walk as the benchmark for every exchange-rate forecast
  - ▶ next chart: the same comparison for EUR/RON

## EUR/RON: ARIMA against the random walk

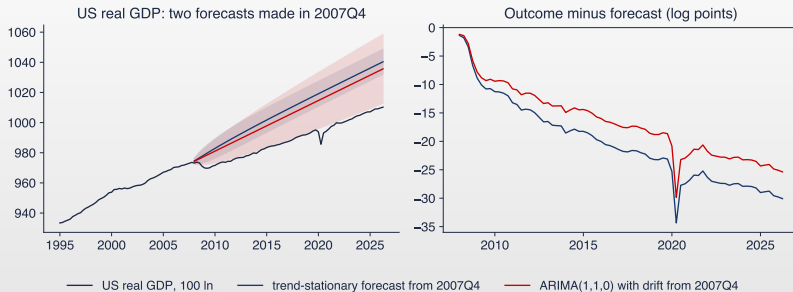


- Parameters estimated on 4 673 daily changes until December 2023; 679 one-day-ahead forecasts from January 2024; right: the cumulated difference of squared errors (below 0: the model beats the random walk)

## Interpreting the EUR/RON comparison

- ▣ RMSE of the daily change (100 ln): random walk 0.1330, with drift 0.1327, ARIMA(1,1,0) 0.1300 ( $\hat{\phi} = 0.169$ )
  - ▶ a gain of 2.2%; over 20 days (33 non-overlapping windows): 0.667 against 0.638, a gain of 4.3%
- ▣ The gain comes from a few days: two jumps of the rate (from 4.9746 to 5.2644 lei over the period) continued on the next day
  - ▶ on calm days the random walk is as good; a formal comparison needs the Diebold–Mariano test ([Chapter 4](#))
- ▣ Lesson of Meese and Rogoff: report the random walk next to any exchange-rate forecast

## A classic series: US real GDP after 2007



- FRED series GDPC1 since 1947; a TS model (linear trend, AR(2) errors) and a DS model (ARIMA(1,1,0) with drift) estimated up to 2007Q4 and forecast to 2026Q2; 95% intervals

## Interpreting the US GDP forecasts

- The TS model expected a return to the pre-2008 line; the gap kept widening:  $-30$  log points at the end (half-width of its interval: 8)
  - ▶ the 2008–2009 loss was never recovered: evidence for a permanent shock, as the DS view of Nelson and Plosser (1982) implies
- The DS model also misses ( $-25$ ), but its interval (half-width 23) at least warns of large uncertainty
  - ▶ both assume the same drift as before 2008; growth after 2008 was slower: a break in the drift (Section 5)
- Unit-root tests on the full series: ADF  $p = 0.82$ , KPSS 0.58:  $I(1)$

## Recap: ARIMA in practice

- ▣ Automatic ARIMA: KPSS for  $d$ , AICc for  $p$ ,  $q$ ; check convergence, roots and residuals
- ▣ AICc compares models with the same  $d$  only
- ▣ Exchange rates: the random walk is a hard benchmark [Meese and Rogoff \(1983\)](#)
- ▣ TS or DS changes the long-run forecast and its interval: US GDP after 2008

## Possible contribution of AI

- ▣ **Code:** a first draft of a script that runs ADF, PP and KPSS on many series and tabulates the verdicts
- ▣ **Explanation:** a second explanation of why the Dickey–Fuller distribution is not Normal, or of a ZA output
- ▣ **Exploration:** unit roots and breaks in all EU inflation rates, or ARIMA benchmarks for many Romanian series
- ▣ Example prompt
  - Write Python code that loads Romanian quarterly real GDP (Eurostat namq\_10\_gdp, seasonally adjusted), runs ADF with constant and trend and KPSS on  $100 \cdot \log$  GDP and on its difference, fits  $\text{ARIMA}(p,1,q)$  with drift for  $p, q \leq 2$ , and plots 8-quarter forecasts with 95% intervals.

## Checks you must run

- ▣ The direction of each test: ADF and PP have the unit root as  $H_0$ , KPSS has stationarity as  $H_0$
- ▣ The deterministic terms, the number of lags and the critical values actually used (Dickey–Fuller, not Normal)
- ▣ That “not rejected” is not reported as “proved”
- ▣ That regressions between trending series are not presented as causal evidence
- ▣ Convergence warnings, roots near the unit circle, residual tests, and AICc compared only for the same  $d$
- ▣ Every cited reference: it must exist; check the DOI

## Key takeaways

- ▣ Trending series are TS (shocks fade) or DS (shocks are permanent); the treatment differs: detrend or difference
- ▣ Regressions between unrelated  $I(1)$  series are spurious: high  $t$  and  $R^2$ , DW near 0
- ▣ ADF and PP test  $H_0$ : unit root with Dickey–Fuller critical values; KPSS tests  $H_0$ : stationarity; use them together
- ▣ The tests have low power near  $\phi = 1$  and are misled by breaks; Zivot–Andrews allows one break
- ▣  $\text{ARIMA}(p, d, q) = \text{ARMA}$  on  $\Delta^d y_t$ ; its forecast intervals grow with the horizon when  $d \geq 1$
- ▣ Automatic selection is a starting point: check  $d$ , convergence, roots, residuals and a random-walk benchmark

## Key formulas

| Quantity           | Formula                                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TS and DS          | $y_t = \alpha + \beta t + u_t, \quad \Delta y_t = \beta + u_t, \quad u_t \sim I(0)$                                                                                       |
| ADF                | $\Delta y_t = c + bt + \gamma y_{t-1} + \sum_{j=1}^k \delta_j \Delta y_{t-j} + \varepsilon_t, \quad H_0: \gamma = 0, \quad \tau = \hat{\gamma} / \text{SE}(\hat{\gamma})$ |
| Lags               | $k_{\max} = 12 (T/100)^{1/4}$ , then AIC or BIC                                                                                                                           |
| KPSS               | $\eta = \sum_{t=1}^T S_t^2 / (T^2 \hat{\lambda}^2), \quad S_t = \sum_{s \leq t} e_s, \quad 5\%: 0.463 (c), 0.146 (c, t)$                                                  |
| ARIMA( $p, d, q$ ) | $\phi(L)(1-L)^d y_t = c + \theta(L)\varepsilon_t$                                                                                                                         |
| Forecast variance  | $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2, \quad \psi(z) = \theta(z) / [\phi(z)(1-z)^d]; \text{ random walk: } h\sigma^2$                                                       |
| ARIMA(0,1,1)       | $\sigma^2 [1 + (h-1)(1+\theta)^2], \quad \alpha_{\text{SES}} = 1 + \theta$                                                                                                |

## Self-assessment

- **Question:** an ADF test with constant and trend gives  $\tau = -2.9$  with  $T = 200$ . Do you reject the unit root at 5%?
  - ▶ **Answer:** no:  $-2.9 > -3.43$ ; with the Normal table you would wrongly reject
- **Question:** ADF does not reject and KPSS does not reject. What do you conclude?
  - ▶ **Answer:** inconclusive: the sample is too short to decide; report both models, or choose  $d$  by the purpose
- **Question:** for ARIMA(0,1,1) with  $\theta = -0.6$  and  $\sigma = 1$ , what is the variance of the 5-step forecast error?
  - ▶ **Answer:**  $1 + 4 \cdot 0.4^2 = 1.64$
- Next: [Chapter 4](#), seasonality: seasonal differences, SARIMA, TBATS and Prophet

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